

Green and Efficient Synthesis of an Adsorbent Fiber by Plasma-induced Grafting of Glycidyl Methacrylate and Its Cd(II) Adsorption Performance

Haonan Chen¹, Mulin Guo², Xiaohong Yao³, Zhengwei Luo⁴, Kai Dong³,
Zhouyang Lian^{3*}, and Wuji Wei^{3*}

¹College of Urban Construction, Nanjing Tech University, Nanjing 210009, China

²College of Chemical Engineering, Nanjing Tech University, Nanjing 210009, China

³College of Environmental Sciences and Engineering, Nanjing Tech University, Nanjing 210009, China

⁴College of Safety Science and Engineering, Nanjing Tech University, Nanjing 210009, China

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Abstract: A new chelating adsorbent for the removal of Cd(II) from aqueous solution, PP-g-GMA-DETA fibers, is prepared by plasma induced grafting of glycidyl methacrylate (GMA) onto the surface of polypropylene (PP) fibers, followed by modification with diethylenetriamine (DETA). The effects of grafting parameters on the grafting degree are investigated. Fourier transform infrared spectroscopy, scanning electron microscopy, and X-ray photoelectron spectroscopy prove the successful grafting of GMA onto the surfaces of PP fibers and the subsequent conversion of epoxy groups of grafted GMA to amine groups, on reaction with DETA. The adsorption performances of Cd(II) by the chelating fibers are evaluated in detail. Kinetic and isothermal parameters are also evaluated. The data fit well with the pseudo-second order and Langmuir isotherm model, respectively. The maximum adsorption capacity of Cd(II) is 46.47 mg g⁻¹ and the prepared fibers show selective adsorbability towards target Cd(II) in presence of competing Mg(II) ions.

Keywords: Polyolefins, Fibers, Adsorption, Grafting, Functionalization of polymers

Introduction

Rapid developments in chemical industries, such as electroplating, metallurgy, chemical engineering, batteries, etc., result in discharge of heavy metal ions, which continuously pollute the environment. These toxic metal ions cannot be biodegraded and tend to accumulate in the human body through food sources, causing serious health problems, even at very low concentrations [1,2]. Amongst the heavy metals, cadmium (Cd) is the most hazardous one and mainly accumulates in the liver, kidneys, and bones, causing pathological changes, ultimately affecting bone growth and metabolism [3]. Therefore, the International Agency for Research on Cancer listed Cd(II) as a human carcinogen, and China listed Cd(II) as the key monitoring indicator for the implementation of total emission control. Thus, the removal of Cd(II) has become a concern that needs to be addressed urgently.

Current technologies to remove Cd(II) ions from water include chemical precipitation, ion exchange, membrane filtration, and adsorption [4-7]. Amongst these, adsorption is one of the most promising techniques, due to its high adsorption efficiency, easy operation, low cost, no secondary pollution, and easy regeneration [8-10]. Use of effective adsorbents in adsorption processes is critical for guaranteeing efficient water treatment. Inorganic adsorbents such as zeolite and diatomite [11,12], organic (polymeric) adsorbents such as cellulose and chitosan [13,14], and carbonaceous

adsorbents such as activated carbon [15], had been developed in the past for the removal of metal contaminants from water resources. However, compared to synthetic fibers, other adsorbents have many drawbacks such as low mass transfer, low adsorption efficiency for low concentrations of heavy metal ions, poor selectivity, high utilization cost, and difficulty in recycling and reuse. It is important to seek new adsorbent materials to rapidly remove Cd(II) ions from water. Polypropylene (PP) fibers as synthetic adsorbent material, has been widely used in different applications such as filtration, oil clean-up, etc., because of its chemical and mechanical stability, larger specific surface area, low cost, and easy availability [16,17]. Although PP fibers have many potential applications, several shortcomings, such as low molecular polarity, poor hydrophilicity, and particularly absence of reactive functional groups, reduce its adsorbability of metal ions from aqueous solutions. According to the Hard and soft acids and bases (HSAB) theory of Pearson, Cd(II) has a strong ability to coordinate with N and O atoms. Therefore, it is very essential to graft functional groups containing N and O atoms onto PP fiber surfaces to improve its hydrophilicity and Cd(II) chelating ability.

In the preparation of functionalized fibers, grafting of various monomers and subsequent modifications to introduce functional groups is a common technique. Glycidyl methacrylate (GMA) is one such useful monomer, which has several advantages such as stability during storage, low price, and almost no toxicity. GMA contains both, acrylate and highly reactive pendant epoxy groups, and the latter can be easily modified into various functional groups, such as amines, sulfonic acid, amidoxime, alcohols, and others [18]. Owing

*Corresponding author: lianzhy@njtech.edu.cn

*Corresponding author: wjwei@njtech.edu.cn

to its excellent metal-ion chelation ability, diethylenetriamine (DETA) has been used as a chelating functional group to modify silica, chitosan, cellulose, and other materials. DETA is ideal for amination, which greatly increases the active adsorption sites in modified materials and also prevents intra- and inter-molecular crosslinking reactions between functional groups [19,20]. Published literature suggests that the functional monomers, GMA and DETA, were used to prepare a chelating adsorbent. Madrid *et al.* [21] prepared PP-g-PGMA functional material by electron beam-induced grafting of GMA onto PP and modifying it with ethylenediamine and iminodiacetic acid. The obtained functionalized materials displayed strong adsorbability towards Cr(VI), Pb(II), and Cd(II). According to Elwakeel *et al.* [19], a functionalized magnetic-macromolecular hybrid material was prepared by grafting GMA onto magnetic chitosan by chemical initiation, followed by amination of the grafted copolymers using diethylenetriamine. The obtained materials were used as sorbents for the removal of Hg(II) ions from aqueous solution. Donia *et al.* [20] successfully prepared a new amine functionalized cellulose for the efficient removal of Cu(II) and Hg(II) from their aqueous solutions. The adsorbent was prepared by polymerization of GMA on the cellulose surface in the presence of hydrophilic *N,N*-methylene bisacrylamide as the cross-linking agent and the coated cellulose was functionalized with DETA. The above methods to introduce functional groups onto the surfaces of substrates by traditional chemically initiated graft polymerizations are complex, time-consuming, and produce a large number of pollutants, which limit the widespread applications of synthetic adsorbents. It is important to seek convenient methods to improve the surface chemistry of polymeric substrates. Compared to chemically initiated grafting, plasma-induced grafting is a surface modification process for PP fibers, which is simple, excellent, and green [22]. It presents unique advantages such as low reaction temperature, less energy consumption. Moreover, grafting can be restricted to the surface of the polymeric matrix, without affecting the bulk properties, due to the lower energy source. To the best of our knowledge, there is still no report on chelation fibers as adsorbent for the removal of Cd(II) ions, prepared by plasma induced grafting of GMA onto the surface of PP fibers, followed by amination of graft copolymers by reaction with DETA.

In this study, we report the design and successful synthesis of novel PP based chelating fibers (PP-g-GMA-DETA) for the removal of Cd(II) ions, which were prepared by plasma induced grafting of GMA onto the surface of PP fibers, followed by modification with DETA. Several characterization methods, such as scanning electron microscopy (SEM), X-ray photoelectron spectroscopy (XPS), and Fourier transform-infrared spectroscopy (FTIR) were employed to investigate the surface morphologies of the unmodified and grafted PP fibers. The effects of grafting parameters on the grafting

degree were investigated. The adsorption behavior of Cd(II) ions, including adsorption kinetics and isotherms, as well as the effect of contact time, pH, and competitive ions was studied. PP-g-GMA-DETA chelating fibers were expected to serve as a new adsorbent material for the treatment of Cd(II) heavy metal ions in wastewater.

Experimental

Materials

PP melt-blown fibers (diameter of 2-5 μm , specific surface area of $1.50 \text{ m}^2 \text{ g}^{-1}$) were prepared in our laboratory. GMA, obtained from Shanghai Adama's Reagent Co. Ltd., was vacuum distilled before use. DETA was purchased from Shanghai Lingfeng Chemical Reagent Co. Ltd. Cadmium nitrate was supplied by Shanghai Jinshan Pavilion New Chemical Reagent Factory. Magnesium chloride hexahydrate, *N,N*-dimethylformamide (DMF), and isopropanol were purchased from Sinopharm Chemical Reagent Co. Ltd. (China). All chemicals used in this study were of analytical grade. Deionized water was used in all experiments.

Preparation of PP-g-GMA-DETA Fibers

The procedure for preparation of PP-g-GMA-DETA fibers can be divided into two stages. Briefly, (1) washed PP fibers (1.0 g) were transferred to a plasma chamber and treated with Ar gas plasma. After irradiation, the fibers were exposed to air for 90 s to obtain PPOOH fibers, which were then immersed in GMA/DMF/water solutions with different concentrations. The grafting polymerization was initiated at the set temperature for a specified time and under the protection of N_2 atmosphere. After grafting, the product was extracted with acetone in a Soxhlet extractor for 10 h to remove the residual monomers and homopolymers. This product was dried in vacuum oven at 35°C and was named PP-g-GMA.

(2) The epoxide groups of PP-g-GMA were converted into amine groups by reaction with DETA. For this, the grafted fibers of PP-g-GMA were soaked in 0.1 mol l^{-1} DETA solution (solvent is a mixture of DMF and water ($v/v=1:1$)) at 80°C for 2 h. The resulting PP-g-GMA-DETA fibers were washed with acetone and deionized water to completely remove residual reactants and dried under vacuum at room temperature till their weight remained constant.

Determination of Grafting Degree

The grafting degree of GMA was determined by a back-titration procedure. Approximately 0.10 g of the extracted PP-g-GMA fibers was dissolved in 100 ml hot xylene and 5 ml trichloroacetic acid (0.3 mol l^{-1}) in xylene was added. The mixture was refluxed for 1 h during which, the reaction between epoxide groups and trichloroacetic acid was complete. The reaction mixture was cooled to room temperature and 100 ml ethanol was added to precipitate the polymer in the

solution and then titrated with 0.05 M NaOH in ethanol using phenolphthalein indicator.

The grafting degree (GR) of PP-g-GMA was calculated by equation (1) as follows:

$$GR = (V_0 - V_1) \times C / w \quad (1)$$

where, V_0 (mL) and V_1 (mL) are the volumes of NaOH solutions, used in the titration of blank test and grafted PP, respectively. Moreover, C (mol l^{-1}) is the molar concentration of NaOH solution and w (g) is the weight of the grafted product.

Characterization

FT-IR spectra of pristine PP and grafted PP fibers were obtained on a Nicolet-Nexus 670 FT-IR spectrometer (Thermo Nicolet Company, USA) over the spectral range of 4000 to 300 cm^{-1} at a resolution of 4 cm^{-1} and 32 scans per sample.

SEM images of pure PP and PP-g-GMA fibers were obtained using a JSM-5900 SEM (JEOL, Japan). These samples were attached to a carbon tape and sputter coated with gold to enhance the electrical conductivity under vacuum, prior to observation.

XPS spectra were obtained to determine the elements on the surface of grafted PP fibers using a PHI-5000 Versa Probe X-ray photoelectron spectroscopic (XPS) instrument (UIVAC-PHI, Japan).

Adsorption Experiment

Cd(II) batch adsorption experiments were performed in glass conical flasks. 1.0 g of each adsorbent was added to 250 mL flask containing 100 mL of adsorbate solution of known concentration. The flask was then placed on an incubator shaker and shaken at 150 rpm for 12 h at 25 °C to ensure adsorption equilibrium. In competitive adsorption experiments, the initial concentrations of Cd(II) were fixed at 100 mg l^{-1} , where as Mg (II) concentrations varied from 0 to 100 mg l^{-1} .

All the adsorption experiments were carried out along with a blank control. Data presented are the average values of triplicate runs and varied by less than 5 % in all the cases studied. The Cd(II) concentration was determined using an inductively coupled plasma-atomic emission spectrometer (ICP-AES, Optima 5300DV, PerkinElmer, USA), and the adsorption capacity (Q) was calculated using the following equation (2):

$$Q = \frac{(C_0 - C_e)V}{W} \quad (2)$$

where, C_0 (mg l^{-1}) and C_e (mg l^{-1}) are the concentrations of Cd(II) before and after the adsorption, respectively. Moreover, V (L) is the volume of Cd(II) aqueous solution, and W (g) is the weight of the fibers added.

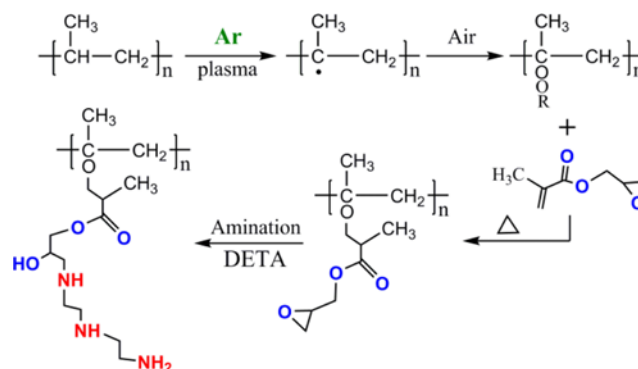
Results and Discussion

Mechanism of Graft-polymerization

Scheme 1 illustrates the mechanism of plasma-induced grafting of GMA on PP fibers and its further modification with DETA. A large number of free radicals were formed on the surface of the fibers when PP fibers were treated with high-energy Ar gas plasma. Free radicals were generated as a result of proton abstraction, C-C bond cleavage, and ion bombardment. Then the treated PP fibers were instantly exposed to atmosphere upon which peroxy, peroxide, and hydroperoxide radicals were instantly formed. These highly reactive groups served as initiators for the “grafting from” polymerization in the presence of unsaturated monomers [23]. The highly reactive groups (peroxy, peroxide, and hydroperoxide) induced the grafting of GMA onto the surface of PP fibers (PP-g-GMA). Subsequently, thermally induced ring-opening reaction of the epoxy groups on the grafted GMA was carried out using DETA as a reagent, which resulted in the bonding of DETA and subsequently the formation of PP-g-GMA-DETA chelating fibers.

Characterization

FT-IR spectroscopy evaluated the chemical structure and surface functional groups of pristine PP fibers and the as-prepared composite fibers. The FT-IR spectra of PP fibers, plasma-treated PP fibers, PP-g-GMA fibers, and PP-g-GMA-DETA fibers are shown in Figure 1. When the spectrum of plasma-treated PP fibers (30 W, 3 min, 15 Pa) (b) was compared with the spectrum of untreated PP fibers (a), a new broad peak appeared at around 3396 cm^{-1} , which corresponded to the peroxide group (-OH deformation). Moreover, the presence of another broad band at approx. 1650 cm^{-1} was seen in the spectrum of plasma-treated PP fibers, which corresponded to the oxygen based groups. The peroxide concentration was determined for plasma-treated PP fibers using the DPPH method, as reported previously [24]. Thus, the plasma-treated PP fibers (30 W, 3 min, 15 Pa)



Scheme 1. Mechanism of plasma-induced grafting of GMA on PP fibers and modification with DETA.

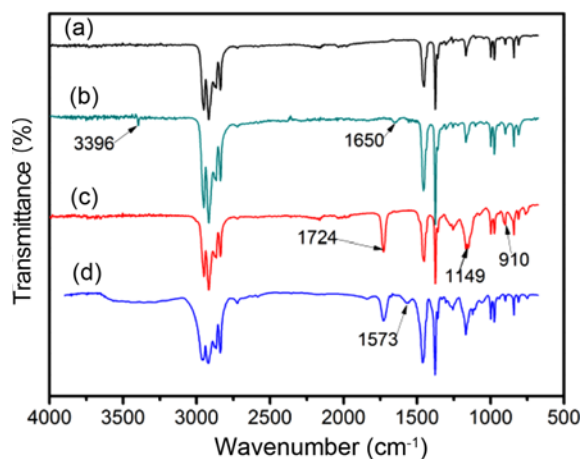


Figure 1. FTIR spectra of fiber samples; (a) PP, (b) plasma-treated PP, (c) PP-g-GMA, and (d) PP-g-GMA-DETA.

were found to have a peroxide concentration of 9.3×10^{-5} mole cm^{-2} . It is obvious that peroxide functionality was formed by the reaction of free radicals (formed by argon plasma) with oxygen and water from the atmosphere. Compared to pristine PP fibers (a), the appearance of a new peak at 1724 cm^{-1} in PP-g-GMA fibers (c) was the result of stretching vibrations of $\text{C}=\text{O}$. The peaks at 1149 cm^{-1} reflected the stretching vibrations of $-\text{C}-\text{O}-$, indicating the presence of the ester group $-\text{COO}-$. A weak absorption peak at around 910 cm^{-1} in pristine PP and plasma-treated PP fibers attributed to alkenyl hydrogen stretching ($-\text{CH}=\text{CH}_2$) in the PP polymer. After grafting with GMA, the intensity of the peak at around 910 cm^{-1} became more intense. This is due to the appearance of a new characteristic epoxy group vibrational peak at 910 cm^{-1} . And the characteristic peaks of epoxy group and alkenyl hydrogen stretching overlapped [25]. These findings indicated that GMA was grafted onto the PP fibers. After amidification, the new band at 1573 cm^{-1} in the spectrum of PP-g-GMA-DETA fibers (d) was assigned to the stretching vibrations of the amide bond ($\text{N}-\text{H}$). In addition, the intensity of the peak at around 910 cm^{-1} was significantly decreased and the final value is similar to that of pristine PP, indicating

that DETA reacted with the epoxy groups [26].

SEM images of PP fibers and PP-g-GMA fibers are shown in Figure 2. Compared to the smooth surface of PP fibers (a), the surface of PP-g-GMA fibers (b) was evidently rough. More specifically, a heterogeneously grafted layer was formed on the surface of the PP-g-GMA fibers, making the fibers coarse. These findings suggested that GMA was grafted onto the fibers during plasma-induced grafting reaction.

XPS analysis of the fibers was carried out to study the chemical changes occurring during the grafting in the top 5–10 nm surface layers, in a better manner. XPS wide scans (survey spectra) of pristine PP, plasma-treated PP fibers, PP-g-GMA-DETA fibers, and PP-g-GMA fibers with GR of $0.34 \text{ mmol} \cdot \text{g}^{-1}$ are presented in Figure 3. The spectrum of pristine PP fibers showed a main peak at 284.6 eV attributed to $\text{C}1\text{s}$, whereas the spectra of plasma-treated PP fibers and PP-g-GMA fibers showed a new peak at 533.1 eV corresponding to $\text{O}1\text{s}$ [27]. Untreated PP fibers showed almost no presence of oxygen on the surface, except for the presence of some atmospheric contamination. When the PP fibers were plasma-treated for 3 min and then exposed to air, the O content increased from about 1.01 % for pristine PP fibers to 5.5 % (See Figure 3). The increase in oxygen content on the polymeric surface after irradiation showed the extent of surface oxidation, due to the combined effects of plasma-treatment and exposure to the atmospheric oxygen. Compared to the survey scan of plasma-treated PP fibers, a significant increase in O content was noted after grafting. The O content increased from 5.5 to 14.23 %, while C content decreased from 94.5 to 85.77 % for copolymers with GR of $0.34 \text{ mmol} \cdot \text{g}^{-1}$. These results suggested that oxygen-containing polar groups were incorporated on the surfaces of PP-g-GMA fibers. After amidification, new peak at 400.5 eV , corresponding to $\text{N}1\text{s}$, appeared in the spectrum of the PP-g-GMA-DETA fibers. This indicated that DETA was grafted onto the PP-g-GMA fibers.

High-resolution XPS analysis of $\text{C}1\text{s}$ peak was carried out to investigate the introduction of principal functional groups onto PP fibers surfaces by plasma-induced grafting. The high-resolution $\text{C}1\text{s}$ spectra of original PP fibers, plasma-treated PP fibers, PP-g-GMA fibers, and PP-g-GMA-DETA

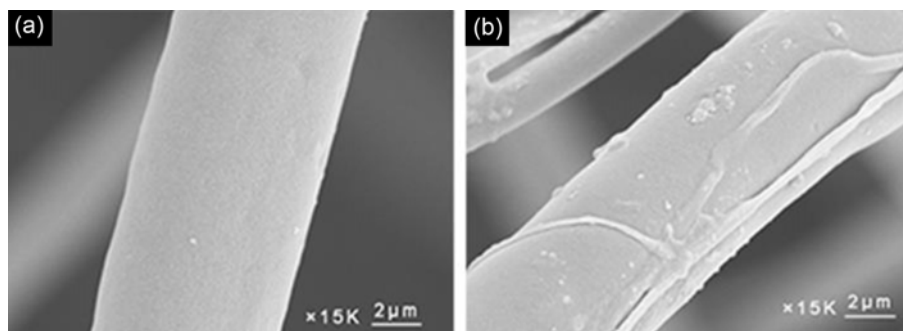


Figure 2. SEM images of PP fibers (a) before and (b) after plasma grafting reaction.

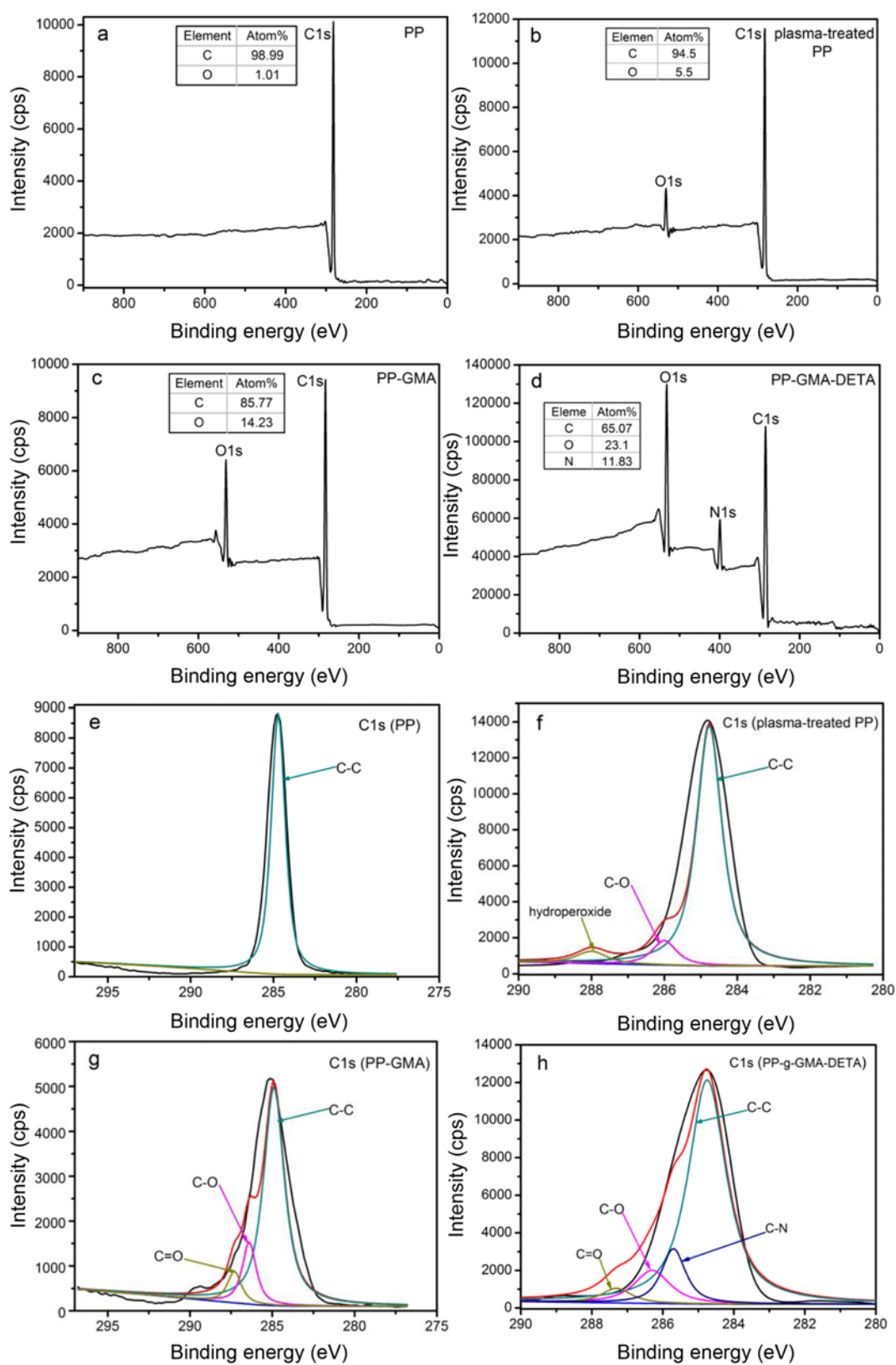


Figure 3. XPS wide scans (survey spectra) (a, b, c, d) and C1s spectra (e, f, g, h) of pristine PP fiber, plasma-treated PP fiber, PP-g-GMA fiber and PP-g-GMA-DETA fiber.

fibers are shown in Figure 3. The spectrum of original PP fibers showed the presence of a peak with binding energy of 284.6 eV for C-C. Interestingly, samples after Ar plasma treatment showed new functional groups, such as C-O and hydroperoxide, corresponding to the binding energies of 286.2 eV and 288.1 eV, respectively. The high-resolution XPS spectrum of PP-g-GMA fibers also showed a peak at 284.6 eV, along with new peaks at 286.4 eV and 287.3 eV, which could be attributed to C-O and C=O, respectively (Figure 3(g)) [27]. These results indicated the successful grafting of GMA onto PP fibers. After amidification, C1s peaks corresponding to C-O (hydroxyl, 286.4 eV), C-N (285.7 eV), and C=O (carboxyl, 287.3 eV) were detected in PP-g-GMA-DETA fibers. This also indicated the successful grafting of DETA onto the PP-g-GMA fibers. It should be mentioned that the results of XPS analysis are in good agreement with the FTIR results.

Effect of the Ar Plasma Parameters on Grafting Degree

Plasma parameters such as treatment time, RF power, and pressure affect the grafting degree, since the rate of formation of free radicals is influenced by plasma treatment. The influence of treatment time, RF power, and pressure on the grafting degree was investigated. Solution parameters of the graft polymerization step were kept constant (GMA concen-

tration of 0.3 (v/v), reaction time of 4 h, and reaction temperature of 95 °C).

The effect of plasma discharge pressure on the grafting degree of GMA is presented in Figure 4(a). It was evident that the grafting degree of GMA increased as the pressure of plasma discharge increased (up to 15 Pa) and then decreased. The maximum grafting degree of GMA (0.34 mmol g^{-1}) was achieved at a pressure of 15 Pa. This could be explained as follows: the plasma discharge created free radical species on the fibers' surface, mainly through polymer chain scission or hydrogen abstraction by bombardment of plasma particles. The density of ionized Ar increased as the pressure of Ar plasma discharge increased, due to which the density of free radical species formed on the fibers' surface increased. These free radical species combined with oxygen from air, forming peroxides on the fibers' surface and facilitated the grafting of GMA. When the pressure of Ar plasma discharge continued to increase beyond a critical value, the probability of collisions between particles in the plasma field increased and the mean free path of particles decreased. So the particles could not absorb the RF power effectively and ionize Ar, resulting in the reduction of high energy particles in the plasma field. This reduced the effect of treatment on the PP fibers surface [28]. So the grafting degree of GMA decreased.

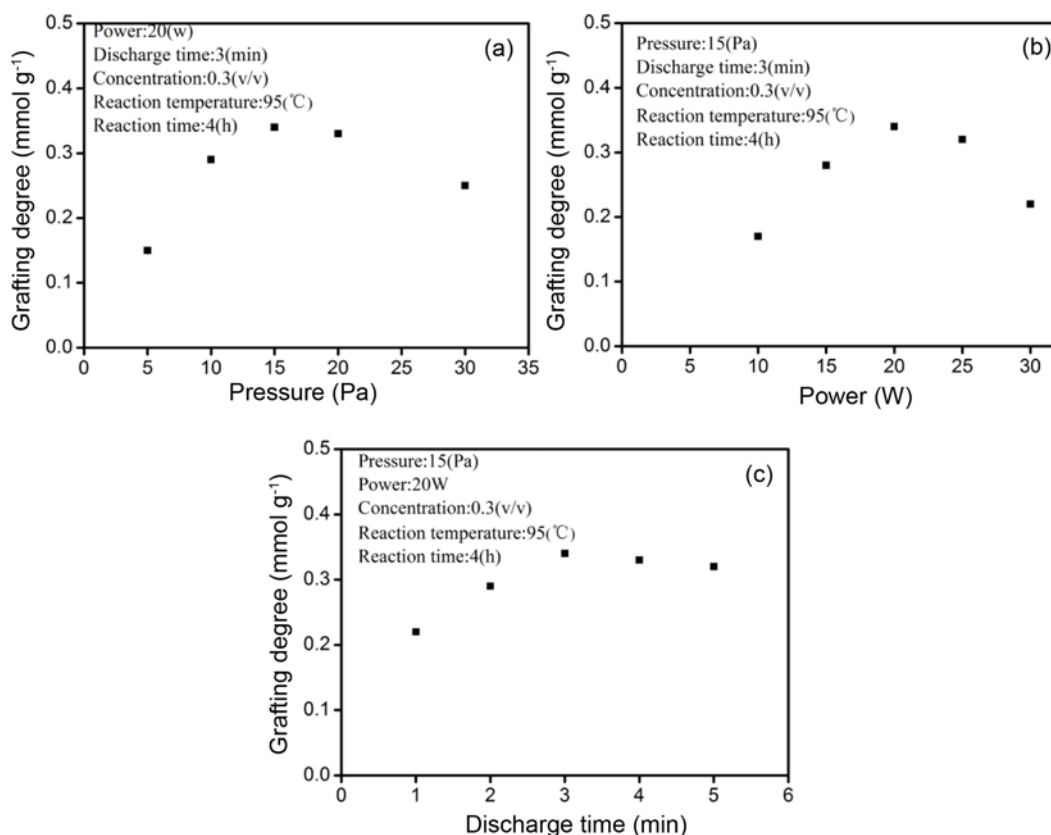


Figure 4. Effects of plasma conditions on the grafting degree of GMA.

Figure 4(b) illustrates the effect of Ar plasma power on the degree of grafting of GMA onto the PP fiber surface. The grafting degree of GMA obviously increased with increase in plasma power up to 20 W and then dropped beyond 20 W. This was due to the fact that an increase in plasma power increased the dissociation degree of plasma. As a consequence, the velocities of the activated particles increased. In other words, the density of the activated particles increased along with their kinetic energies. These activated particles could produce more number of free radicals on the PP fibers' surface. However, increased power (above 20 W) increased the etching (the breaking up of bonds and chain scission) of polypropylene surface, due to the bombardment of PP fibers' surface with high energy plasma particles [29]. This decreased the grafting degree of GMA.

The effect of treatment time of Ar plasma on the grafting degree of GMA is shown in Figure 4(c). It was found that the grafting degree increased initially with treatment time, passed through a maximum at 3 min, and then leveled off. This may be due to enhanced etching beyond the central level of treatment time (3 min), the effect of which dominated the functionalization effect. Moreover, prolonged etching (beyond 3 min) resulted in the removal of free radicals attached to the PP fibers' surface and thereby decreased the graft rate of GMA [30].

To summarize, Ar plasma discharge pressure of 15 Pa,

plasma power of 30 W and plasma treatment time of 3 min produced optimal grafting degree of GMA.

Effect of the Graft Polymerization Parameters on Grafting Degree

The free radicals eventually remained on the surfaces of the fibers after the plasma treatment, which reacted with oxygen from the atmosphere and generated peroxides on the PP fibers' surface. These peroxides later formed macro-radicals, which initiated the grafting process upon heating. The influence of monomer concentration, reaction temperature, and reaction time on the grafting degree was investigated. Ar plasma parameters for the graft polymerization step were kept constant (Ar plasma discharge pressure of 15 Pa, plasma power of 30 W, and plasma treatment time of 3 min).

Figure 5(a) shows the effect of GMA concentration on the grafting degree, under optimized plasma conditions. The grafting degree increased dramatically till a concentration of 0.3 (v/v) GMA, and then dropped suddenly. In a specific range of concentration, higher grafting rates were expected with increase in monomer concentration since at any instant, peroxides generated on the PP fibers surface were able to interact with monomer molecules. However, at monomer concentration above 0.3 (v/v), there was a significant increase in the formation of homopolymer as well. This was expected because viscosity increased with increase in monomer

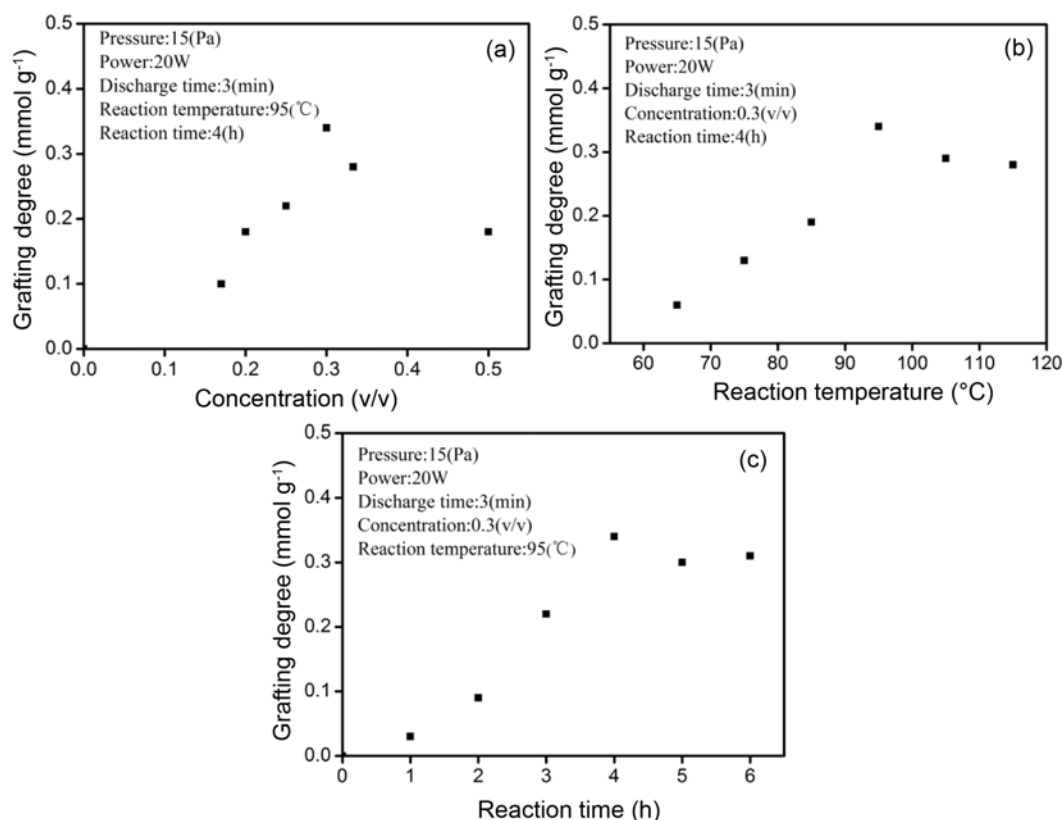


Figure 5. Effects of liquid phase grafting reaction conditions on the grafting degree of GMA.

concentration. The grafting solution turned very viscous, as a result of which the diffusion of monomers onto the PP fibers' surface was hindered, and as a consequence the homopolymer formation increased [31].

Generally, temperature increases the kinetic energy of molecules and promotes polymerization up to a certain limit. Hence, two competing reactions, graft polymerization and homopolymerization, are stimulated at the same time. Figure 5(b) shows the effect of temperature on the grafting degree under optimized conditions, according to which clearly almost no grafting occurred below 60 °C. However, in the temperature range of 65-95 °C, the grafting degree increased considerably, which was maximal at 95 °C. The initial rise in the grafting degree was not only due to increased molecular collisions, but also due to additional free radicals generated during the thermal decomposition of peroxides and hydroperoxides, on the backbone of the PP fibers [32]. Its subsequent decrease could be attributed to homopolymerization reaction at 95 °C, which leaves less amount of GMA monomer to react with the PP fibers' surface.

Variations in the grafting degree with reaction time are shown in Figure 5(c). The grafting degree gradually increased with time, but eventually stabilized after 4 h. This implied that the graft polymerization was at equilibrium, and further extension of the reaction time no longer increased the yield. This behavior could be due to the rapid consumption of monomer at low concentration, and with time, less monomer and free radical peroxides were available for the grafting reaction [33].

From these experiments, it was evident that 0.3 (v/v) aqueous GMA, 95 °C, and 4 h were the best reaction conditions for the preparation of PP-g-GMA.

Adsorption Performance of Cd(II) on PP-g-GMA-DETA

The PP-g-GMA-DETA could be used as chelating fibers for the adsorption of metal ions. The lone pair of electrons on nitrogen atoms of the graft chains (amino groups) could form coordination bonds with the empty outer electronic orbitals of metal ions. The nitrogen atoms had good ability to coordinate with Cd(II) [34]. The Cd(II) adsorption performance was the crucial aspect to evaluate the prepared materials.

Effect of pH

It is well known that solution pH is an important parameter affecting the adsorption of Cd(II) ions on adsorbents. This is because it not only affects the formation of Cd(II) ions in solution, but also influences the surface properties of the adsorbents in terms of dissociation of functional groups and surface charges. The uptake values were recorded prior to the precipitation limit for Cd(II) ions. Figure 6 shows the performances of PP, PP-g-GMA, and PP-g-GMA-DETA fibers in the adsorption of Cd(II) ions at different solution pH values. Negligible Cd(II) sorption by the PP fibers, was due to poor hydrophilicity of original PP fibers and also the

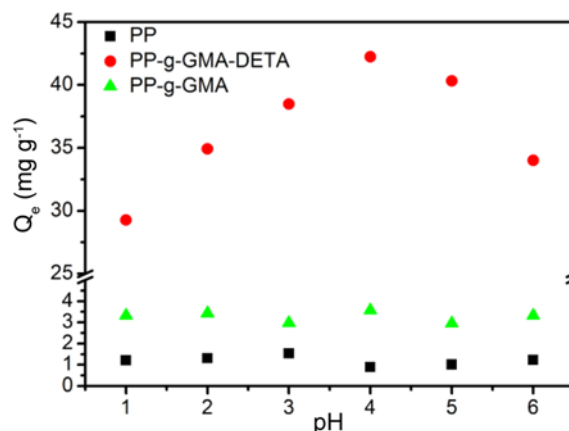


Figure 6. Effect of pH on Cd(II) ion adsorption on PP-g-GMA-DETA, PP-g-GMA and PP fibers.

absence of any reactive functional groups. PP-g-GMA fibers showed lesser Cd(II) sorption than PP-g-GMA-DETA fibers in the pH range studied. This could be explained by the two following facts: first, the PP-g-GMA fibers were hydrophobic, which reduced the adsorption of Cd(II) from water. Second, PP-g-GMA-DETA fibers were modified with hydrophilic groups (amino groups), which improved the hydrophilicity and Cd(II) chelating ability of PP-g-GMA-DETA. The hydrophilic surfaces of PP-g-GMA-DETA fibers not only increased its compatibility with the heavy-metal ions in solution, but also enhanced its adsorption rate and their capacity for Cd(II) adsorption. In contrast to the PP-g-GMA fibers that could hardly adsorb Cd(II) ions in the pH range studied, the PP-g-GMA-DETA adsorbent showed a pH-dependent adsorption behavior and the adsorption of Cd(II) ions increased significantly with increase in pH in the pH range of 1.0-4.0. The maximum value was reached at pH 4.0, which then declined gradually. This phenomenon could be attributed to several aspects. Firstly, the amine groups of DETA grafted on the PP-g-GMA-DETA adsorbent surface were the functional groups responsible for Cd(II) ion adsorption. At low pH values (high acidic environment), a relatively high concentration of protons strongly competed with the Cd(II) ions for the amine sites, and thus the adsorption of Cd(II) ions was significantly reduced. On the other hand, the protonation of amine groups led to a strong electrostatic repulsion with the Cd(II) ions to be adsorbed. As a result, it became difficult for the Cd(II) ions to establish contact with the surface of the adsorbent for adsorption to occur and thus showed poor adsorption performance at low pH values. With increase in pH values (low proton concentrations), the competition of protons with the Cd(II) ions for the amine groups became less significant. Additionally more number of the amine groups were present in their neutral form, which reduced the electrostatic repulsion with the Cd(II) ions. Hence, more number of Cd(II) ions could be

adsorbed on the surfaces of PP-g-GMA-DETA fibers. This explained the increase in Cd(II) ion adsorption on the PP-g-GMA-DETA adsorbent surface at higher pH values. At pH value above 4.0, Cd(II) easily hydrolyzed in solution, which resulted in the decrease in adsorption capacity [34].

Adsorption Kinetics

Knowledge of adsorption of an adsorption system is necessary for the development and evaluation of adsorbents, since it helps to determine the phenomenological coefficients that characterize the transport of adsorbate within adsorbents. Figure 7 shows the adsorption kinetics of Cd(II) on PP-g-GMA-DETA adsorbent. The experimental results show that the adsorption of Cd(II) by the adsorbent was rapid before 120 min, steady between 120-180 min, and reached the adsorption equilibrium at 180 min, with maximum adsorption capacity of 41.87 mg g⁻¹. This phenomenon could be explained as follows: PP-g-GMA-DETA chelating fibers had a large surface area with most of its reactive groups grafted on the surface of the fibers. Hence, the adsorption rate of Cd(II) on the adsorbent mainly depended on the transfer of Cd(II) from the solution to the adsorption sites on fibers' surface and the number of available adsorption sites. The initial section of the curve corresponded to the greater availability of adsorption sites and a larger concentration gradient between the solution and the fibers' surface. These two conditions could explain the fast initial accumulation of Cd(II). The adsorption mainly occurred at the reactive sites covering the surface of the adsorbent. The second step, much slower, was controlled by the decrease in concentration gradient and available adsorption sites.

The experimental data were modeled based on the pseudo-first- and pseudo-second-order models in order to analyze the adsorption characteristics of the metal ions on the adsorbent. The model equations are expressed as equations (3) and (4) [35,36].

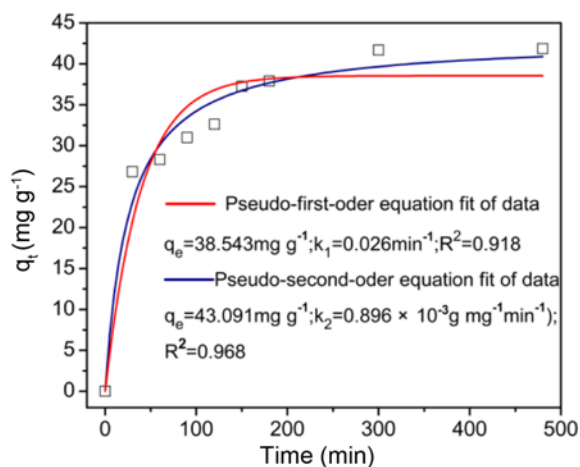


Figure 7. Curves for adsorption kinetics of Cd(II) ion on PP-g-GMA-DETA.

Table 1. Comparison of Cd(II) semi-saturation time of various adsorbents

Adsorbent	Conditions	$t_{1/2}$	Ref.
Corn straw biochar	T=293 K; pH=6	72.7 min	[38]
p(H/V)-T hydrogel	T=298 K; pH=5	70.9 min	[39]
CMC-LF fiber	T=298 K; pH=6	58.2 min	[40]
MSB	T=298 K; pH=5	41.2 min	[41]
PP-g-GMA-DETA fibers	T=298 K; pH=4	25.9 min	This study

The pseudo-first-order model follows:

$$q_t = q_e(1 - e^{-k_1 t}) \quad (3)$$

The pseudo-second-order model follows:

$$q_t = \frac{q_e^2 k_2 t}{(1 + q_e k_2 t)} \quad (4)$$

where q_e is the amount of Cd (II) adsorbed at equilibrium (mg g⁻¹); q_t is the amount adsorbed at time t , (mg g⁻¹); k_1 is the first-order equilibrium rate constant (1 min⁻¹), and k_2 is the second-order equilibrium rate constant (g mg⁻¹ min⁻¹).

The fitting of equations (3) and (4) to the experimental data in Figure 7 was conducted and the model parameters from the calculation are also given in Figure 7. It was obvious that the correlation coefficients (R^2) for the pseudo-second-order kinetic model had higher values than those for the pseudo-first-order kinetic model, and the theoretical adsorption capacities (q_e) calculated from the pseudo-second-order kinetic equation were much closer to the experimental adsorption capacities (q_m). This demonstrated that the pseudo-second-order adsorption model predominantly depicted the adsorption process. It also confirmed that the adsorption process was probably dominated by a chemical adsorption phenomenon and that chelation was the main adsorption mechanism of adsorption [37].

The semi-saturation time describes the rate of adsorption by the adsorbent. In Table 1, the semi-saturation time of Cd(II) on PP-g-GMA-DETA chelating fibers was comparable to other adsorbents reported for cadmium ions [38-41], which indicated that the adsorption rate of PP-g-GMA-DETA adsorbent was very fast, probably since the adsorption occurred on the surface of chelating fibers and the mass transfer was fast.

Adsorption Isotherms

Adsorption isotherms examine how adsorbates interact with adsorbents and how adsorption uptakes vary with adsorbate concentrations at given pH values and temperatures. The isotherms for adsorption of Cd(II) onto PP-g-GMA-DETA chelating fibers were studied at pH 4.0 and temperature of 298K. Results in Figure 8 show that initially the isotherm increased sharply as the equilibrium concentrations increased, indicating that many readily accessible sites were available for adsorption. However, as the concentration increased to

30 mg l⁻¹, a plateau was reached, indicating that no more sites were available for adsorption.

Two equations most commonly used to describe the isothermal adsorption of solid-liquid adsorption system are Langmuir equation and Freundlich equation. The Langmuir equation indicated that the adsorption of the adsorbent was a single layer, but Freundlich equation was for the multilayer adsorption. Considering the linear form of these two isotherms [42,43], equations can be written as follows:

$$\text{Langmuir: } \frac{C_e}{Q_e} = \frac{C_e}{q_m} + \frac{1}{K_L q_m} \quad (5)$$

$$\text{Freundlich: } \ln Q_e = \ln K_F + \frac{1}{n} \ln C_e \quad (6)$$

where Q_e (mg g⁻¹) is the adsorption capacity at equilibrium, C_e (mg l⁻¹) is the equilibrium concentration of Cd (II) ions in solution, q_m (mg g⁻¹) is the saturated adsorption capacity, K_L

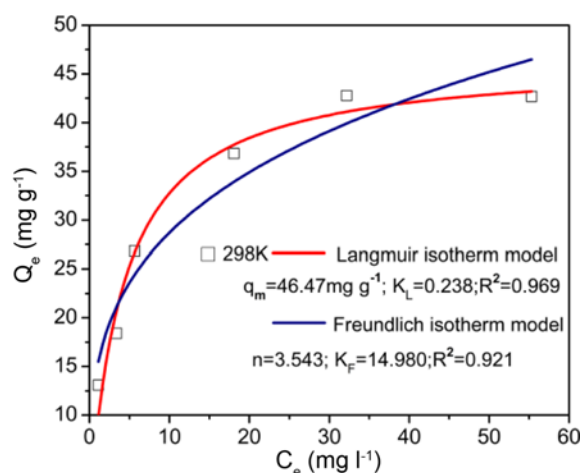


Figure 8. Adsorption isotherms of Cd(II) on fibrous adsorbents at 298 K.

Table 2. Adsorption capacities for Cd(II) on various adsorbents reported in the Literature

Adsorbent	Conditions	Adsorption capacity	Ref.
Porous resin	T=298 K; pH=7	3.51 mg g ⁻¹	[44]
Aminated lignin	T=293 K; pH=6	7.1 mg g ⁻¹	[45]
Attapulgite	T=298 K; pH=5.45	10.38 mg g ⁻¹	[46]
Steel slag-montmorillonite	T=298 K; pH=6-7	12.45 mg g ⁻¹	[47]
Poly(HEMA)-CB cryogel disks	T=298 K; pH=7	25.5 mg g ⁻¹	[48]
EDTA-EDA-PAN nanofibers	T=298 K; pH=6	32.68 mg g ⁻¹	[49]
Activated carbon fiber	T=298 K; pH=5.6	33.8 mg g ⁻¹	[50]
PP-g-GMA-DETA fibers	T=298 K; pH=4	46.47 mg g ⁻¹	This study

is an empirical parameter, K_F is the binding energy constant, and n is the Freundlich constant.

The isothermal adsorption models of Langmuir and Freundlich were fitted with experimental data, and the results were analyzed. The results are also presented in Figure 8. For the two studied models, the Langmuir equation showed a more significant correlation ($R^2 > 0.969$) than the Freundlich equation with the experimental data, which suggested monolayer adsorption of Cd (II) on PP-g-GMA-DETA.

The adsorption capacity and performance were compared with other adsorbents for Cd(II) removal reported in the literature [44-50]. Table 2 showed that the removal efficiency of Cd(II) using PP-g-GMA-DETA chelating fibers was comparable with the other adsorbents reported for cadmium ions. The high sorption capacity of the PP-g-GMA-DETA chelating fibers towards Cd(II) ions showed the promising practical application of this adsorbent for removal of Cd(II) ions from wastewater.

Effect of Mg(II) on Adsorption

In removal of heavy metals from multi-metal solution by sorbent materials, the co-existence of cations could reduce the adsorption capacity through a competing effect. Since the metal ion Mg(II) was present widely in natural water and wastewater, it was important to investigate the effect of Mg(II) on Cd(II) adsorption. The results are presented in Figure 9. With Mg(II) initial concentration increasing from 0 to 100 mg l⁻¹, the Cd(II) adsorption capacity of PP-g-GMA-DETA fibers remained almost unchanged, which was about 43 mg g⁻¹. Thus, the presence of Mg(II) in this concentration range did not reduce the adsorption capacity for Cd(II) markedly. This phenomenon could be explained by the fact that the amino/amide groups modified on PP-g-GMA-DETA could chelate with Cd(II), whereas the interaction between Mg(II) and PP-g-GMA-DETA was weaker than that between Cd(II) and PP-g-GMA-DETA [34]. This makes PP-g-GMA-DETA fibers potential candidates for selective removal of

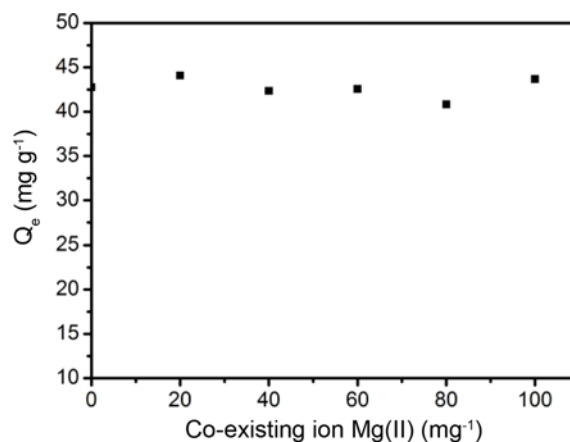


Figure 9. Influence of presence of Mg(II) ions on the adsorption capacity of Cd(II).

Cd(II) ions from wastewater.

Conclusion

The PP-g-GMA-DETA fibers with chelating amine moieties, which could remove Cd(II) ions rapidly, were prepared in two steps. PP fibers were grafted with GMA by plasma treatment and then modified with DETA. The reaction was controlled by plasma parameters and graft polymerization conditions to prevent homopolymerization and increase grafting efficiency. Experimental results have shown that the optimal plasma conditions included Ar pressure of 15 Pa, power of 20 W, and time of 3 min. The optimal grafting reaction conditions were a GMA concentration of 0.3(v/v), time of 4 h, and temperature of 95 °C. FT-IR, SEM, and XPS proved the grafting of GMA onto the surface of PP fibers and subsequent conversion of epoxy groups of grafted GMA to amine groups by reaction with DETA.

The features of the adsorbent for adsorption of Cd(II) ions were investigated. Experimental results showed that PP-g-GMA-DETA fibers had excellent adsorption performance for Cd(II), and the sorption capacity was maximum at pH 4. The adsorption kinetics closely followed the pseudo-second-order kinetic model, indicating the importance of chemical adsorption in the process and showing chelation as the main adsorption mechanism. The experimental adsorption isothermal data was best fitted with the Langmuir model, which suggested a monolayer adsorption of Cd(II) on PP-g-GMA-DETA. The maximum adsorption capacity was found to be 46.47 mg g⁻¹ at pH 4. Moreover, the chelating fibers had the ability to selectively adsorb Cd(II) from solution, which contained competing Mg(II) ions. Furthermore, the PP-g-GMA-DETA fibers showed good chemical and mechanical stability and their preparation was fast, economical, and environmentally friendly. Thus, the chelating fibers can be potentially applied for the efficient and rapid removal of Cd(II) from wastewater.

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